

自然语言处理 (2025 春季学期)

十四、预训练语言模型的延伸



大纲

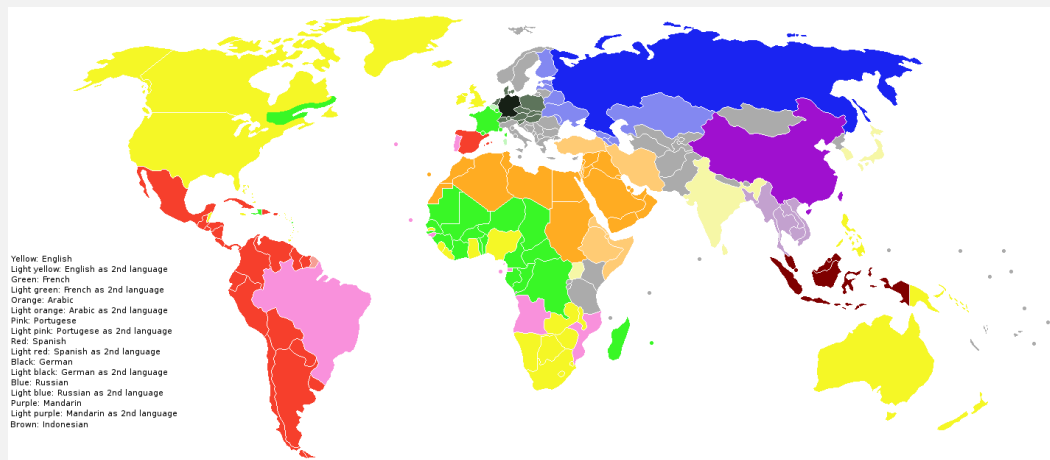
- 多语言预训练模型
- 代码预训练模型
- 多模态预训练模型
- 具身预训练模型



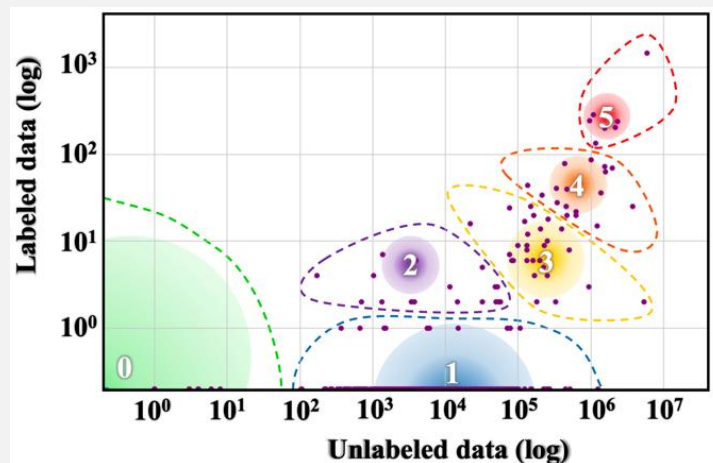


背景

□ 世界上有超过7,000种语言，除少数语言外，绝大部分语言仅存在少量甚至不存在任何标注数据



https://commons.wikimedia.org/wiki/File:Main_world_languages.png



[Joshi et al. \(2020\).](#)



背景

□ 数据分布的长尾现象（Long Tail Distribution）



Figure credit: Graham Neubig



多语言BERT

□ Multilingual BERT

- <https://github.com/google-research/bert/blob/master/multilingual.md>
- 统一的多语言表示空间（104种语言）

```
>>> from transformers import pipeline
>>> unmasker = pipeline('fill-mask', model='bert-base-multilingual-cased')
>>> output = unmasker('我like[MASK]')
>>> pprint(output)
[{'sequence': '[CLS] 我 like 你 [SEP]',
  'score': 0.10890847444534302,
```

- 为什么有效？
 - 语言的混合使用（Code-Switch现象）
 - 共享子词

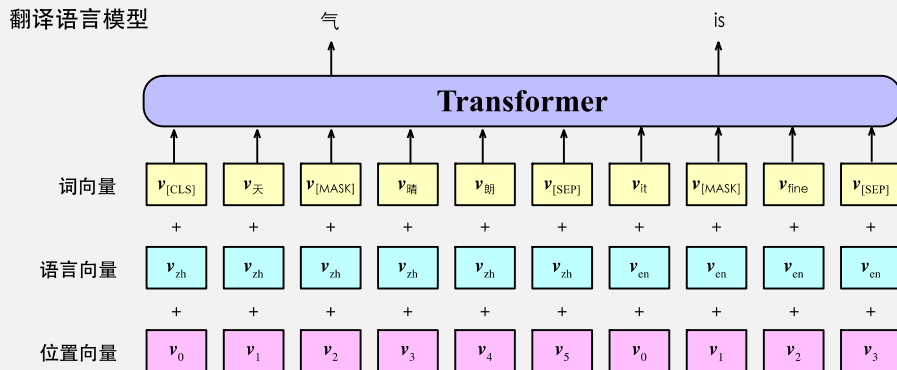




跨语言预训练语言模型

□ XLM (Lample and Conneau, NeurIPS, 2019)

— 翻译语言模型 (Translation Language Modeling, TLM)



— 依赖双语平行句对

- 大规模平行句对获取难度较高
- 受限于句子级上下文 (篇章/文档级平行数据更为稀少)





□ 零样本迁移（Zero-shot Transfer）

- 将源语言（资源丰富，如英语）上训练得到的模型直接应用于目标语言（通常为资源稀缺语言）
- XTREME基准测试集（Hu et al., ICML 2020）

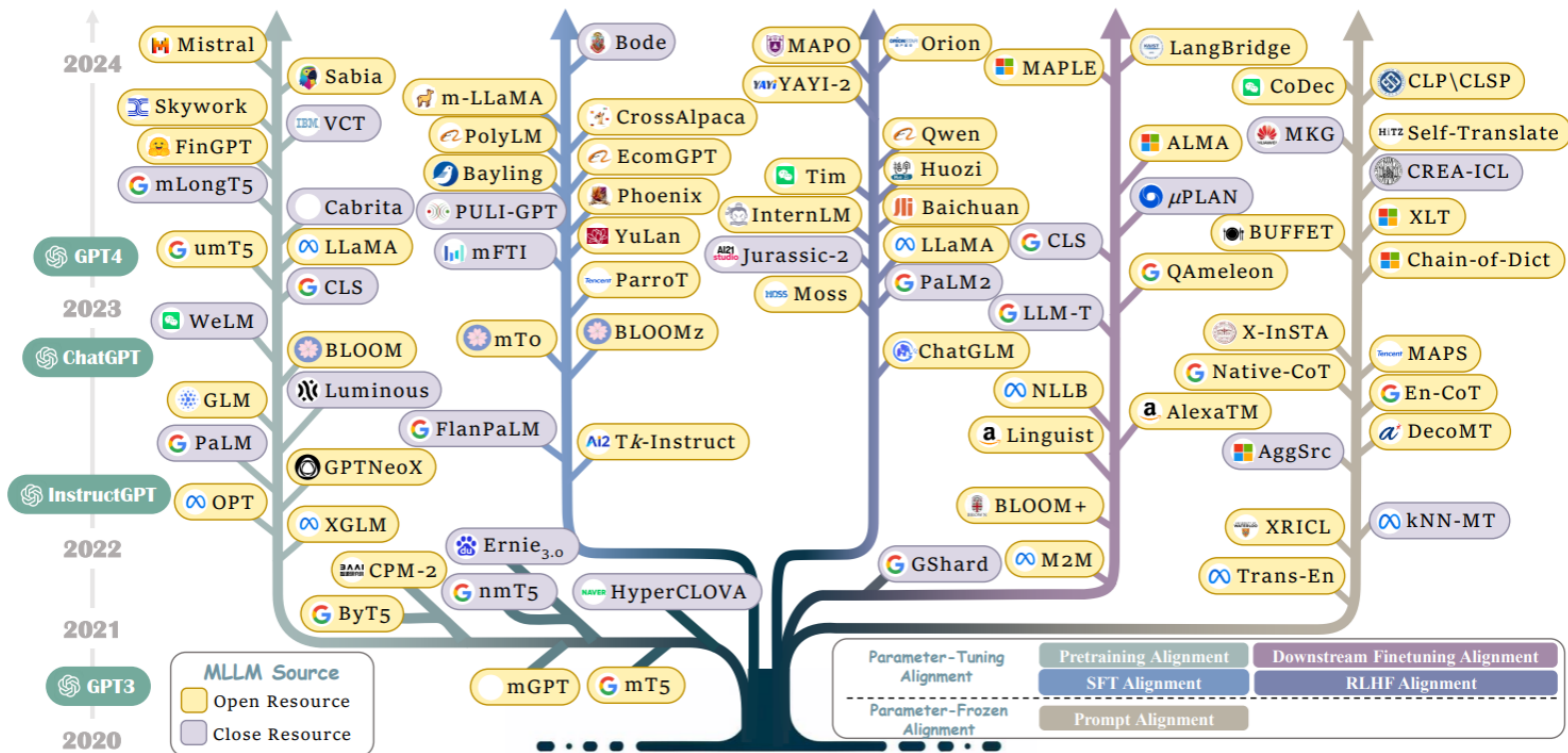
表 9-1 XTREME 数据集的相关信息

任务类型	语料库	数据集规模（训练/开发/测试）	测试集来源	语言数	任务描述
分类	XNLI	392,702/2,490/5,010	翻译	15	文本蕴含
	PAWS-X	49,401/2,000/2,000	翻译	7	复述识别
结构预测	POS	21,253/3,974/47-20,436	独立标注	33	词性标注
	NER	20,000/10,000/1,000-10,000	独立标注	40	命名实体识别
问答	XQuAD	87,599/34,726/1,190	翻译	11	片段抽取
	MLQA	87,599/34,726/4,517-11,590	翻译	7	片段抽取
	TyDiQA-GoldP	3,696/634/323-2,719	独立标注	9	片段抽取
检索	BUCC	–/–/1,896-14,330	–	5	句子检索
	Tatoeba	–/–/1,000	–	33	句子检索





多语言大模型的进展





大纲

- 多语言预训练模型
- 代码预训练模型
- 多模态预训练模型
- 具身预训练模型





自然语言模型中加入编程语言的意义

□ 从 NL 到 PL 的角度

- 掌握编程语言，辅助程序员进行代码开发，提高生产力
- GitHub Copilot 将开发人员的生产力提高 26.08%，经验较少的人显示出更大的提升^[1]

□ 从 PL 到 NL 的角度

```
for ($i=0; $i < 10; $i++) {  
    echo $i;  
}  
  
$fruits = array("apple", "banana", "orange");  
  
foreach ($fruits as $fruit) {  
    echo $fruit;  
}
```

代码蕴含大量结构化知识

```
def solve_chicken_rabbit(total_heads, total_legs):  
    # 鸡兔总数最大值为总头数  
    max_chicken_rabbit = total_heads  
  
    for chicken_count in range(max_chicken_rabbit + 1):  
        rabbit_count = total_heads - chicken_count  
        if (chicken_count * 2) + (rabbit_count * 4) == total_legs:  
            return chicken_count, rabbit_count  
  
    # 无解情况返回None  
    return None
```

增强大模型的推理能力

数据集	The Stack v1	The Stack v2
初始大小	6.4TB	67.5TB
清洗过后	2.9TB	32.1TB
#Token	~200B	~900B
语言数量	358	658

代码数据规模巨大

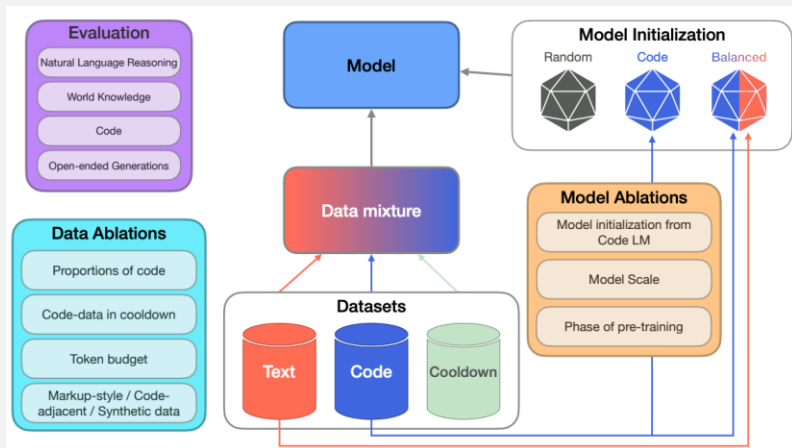




自然语言模型中加入编程语言的意义

□ 在预训练数据中混入代码数据显著增强模型能力

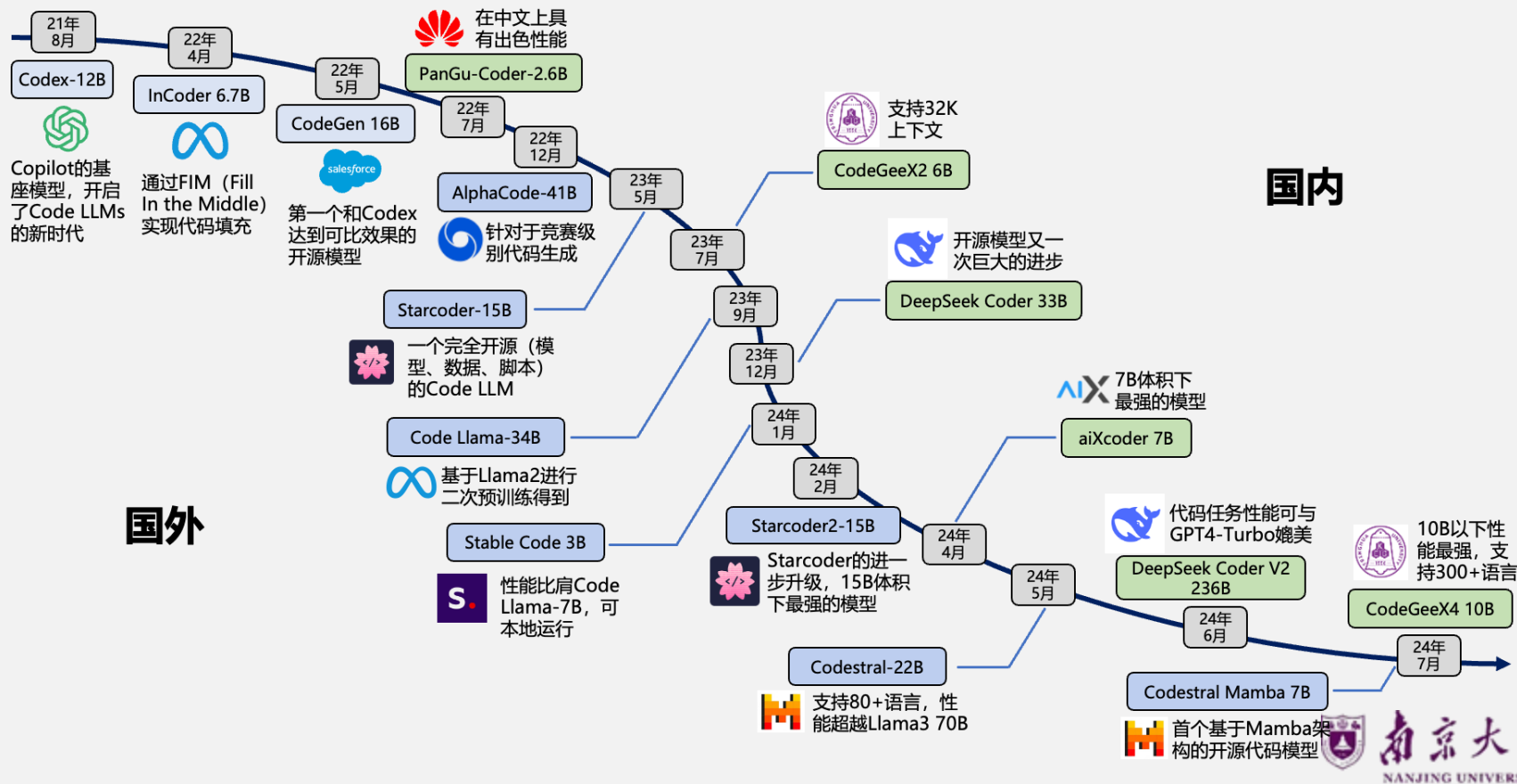
- 自然语言推理相对提高 8.2%
- 世界知识相对提高 4.2%
- 代码性能提高 12 倍



Aryabumi, Viraat, et al. To Code, or Not To Code? Exploring Impact of Code in Pre-training. *arXiv preprint arXiv:2408.10914* (2024).



代表性代码大模型





代码的特性

结构性

- 递归
- 层次化
- 条件分支

编程语言

```
def fib(n):  
    if n == 1 or n == 2:  
        return 1  
    return fib(n-1) + fib(n-2)
```

自然语言

计算斐波那契数列的第n个元素

可执行性

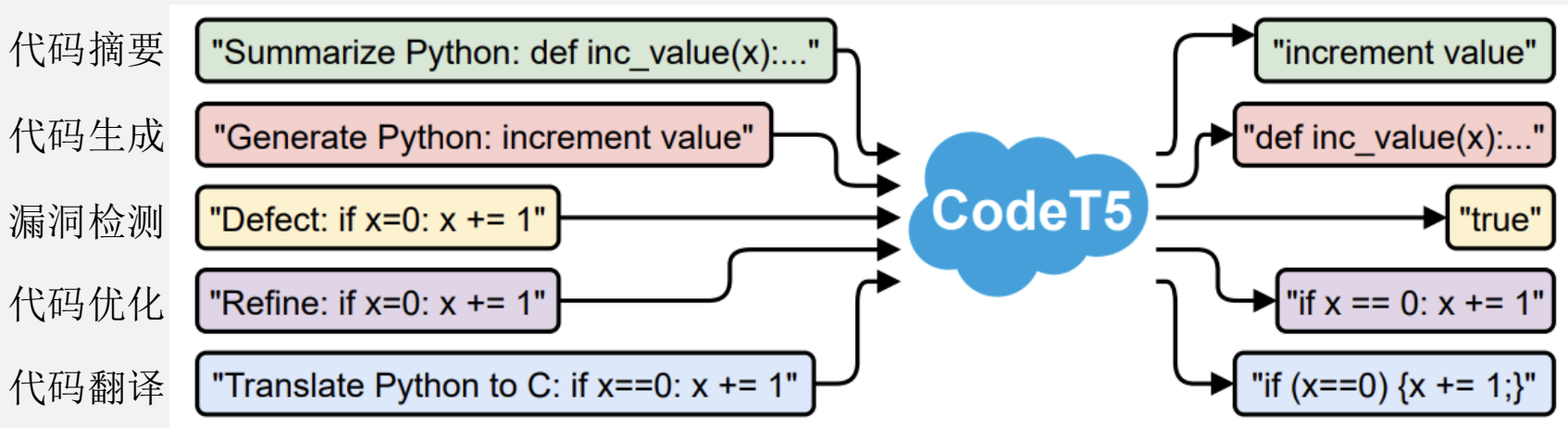
Input	Output
1	1
2	1
3	2
...	...





面向代码的应用

面向代码的应用：以代码的生成与理解为目标，提升编程效率



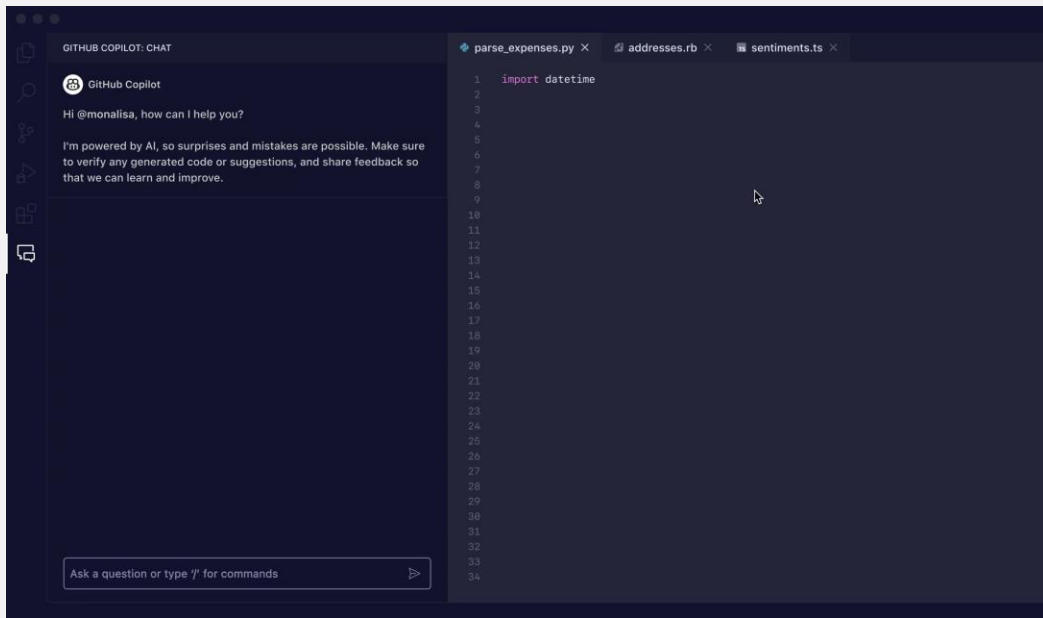
Yue Wang, Weishi Wang, Shafiq Joty, and Steven C.H. Hoi. 2021.

[CodeT5: Identifier-aware Unified Pre-trained Encoder-Decoder Models for Code Understanding and Generation](#). ACL 2021.



代码大模型 (Code LLMs)

- 类似的技术应用于编程语言 (PL)，能够理解和生成代码
 - 根据指令 (instruction/prompts) 执行各项代码相关的任务
 - 代码生成、修复、翻译、摘要.....





基于代码的应用

基于代码的应用：以代码的结构或可执行性为手段，完成目标任务

Question: In Fibonacci sequence, it follows the rule that each number is equal to the sum of the preceding two numbers. Assuming the first two numbers are 0 and 1, what is the 50th number in Fibonacci sequence?

The first number is 0, the second number is 1, therefore, the third number is 0+1=1. The fourth number is 1+1=2. The fifth number is 1+2=3. The sixth number is 2+3=5. The seventh number is 3+5=8. The eighth number is 5+8=13. (Skip 1000 tokens)
The 50th number is 32,432,268,459.

CoT

```
length_of_fibonacci_sequence = 50
fibonacci_sequence = np.zeros(length_of_)
fibonacci_sequence[0] = 0
fibonacci_sequence[1] = 1
for i in range(3, length_of_fibonacci_sequence):
    fibonacci_sequence[i] = fibonacci_sequence[i-1] + fibonacci_sequence[i-2]
ans = fibonacci_sequence[-1]
```

PoT

32,432,268,459



12,586,269,025



Question: Ketty saves 20000 dollars to the bank. After three years, the sum with compound interest rate is 1000 dollars more than the sum with simple interest rate. What is the interest rate of the bank?

Assuming the interest rate is x . The sum after two years with simple interest rate is $20000 + x * 20000 * 3 = 20000 + 60000x$. The sum after two years with compound interest rate is $20000 * (1 + x)^3 = 200000 + 60000 * x + 60000x^2 + 20000x^3$. The difference can be written as $60000x^2 + 20000x^3 = 1000$. In order to solve x , we can use the quadratic formula. $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$, ..., $x = (-20000 \pm \sqrt{6160}) / 120000$, $x = -0.051333$.

CoT

```
interest_rate = Symbol('x')
sum_in_two_years_with_simple_interest = 20000 +
interest_rate * 20000 * 3
sum_in_two_years_with_compound_interest = 20000 * (1 +
interest_rate)**3
# Since compound interest is 1000 more than simple interest.
ans = solve(sum_after_in_yeras_with_compound_interest -
sum_after_two_years_in_compound_interest - 1000,
interest_rate)
```

PoT

-0.051333



$x = 0.24814$



```
# Q: 2015 is coming in 36 hours. What is the date one week from today in MM/DD/YYYY?
# If 2015 is coming in 36 hours, then today is 36 hours before.
today = datetime(2015, 1, 1) - relativedelta(hours=36)
# One week from today,
one_week_from_today = today + relativedelta(weeks=1)
# The answer formatted with %m/%d/%Y is
one_week_from_today.strftime('%m/%d/%Y')
```

Q: On the table, you see a bunch of objects arranged in a row: a purple paperclip, a pink stress ball, a brown keychain, a green scrunchiephone charger, a mauve fidget spinner, and a burgundy pen. What is the color of the object directly to the right of the stress ball?

```
# Put objects into a list to record ordering
objects = []
objects += [['paperclip', 'purple']] * 1
objects += [['stress ball', 'pink']] * 1
objects += [['keychain', 'brown']] * 1
objects += [['scrunchiephone charger', 'green']] * 1
objects += [['fidget spinner', 'mauve']] * 1
objects += [['pen', 'burgundy']] * 1
# Find the index of the stress ball
stress_ball_idx = None
for i, object in enumerate(objects):
    if object[0] == 'stress ball':
        stress_ball_idx = i
        break
# Find the directly right object
direct_right = objects[stress_ball_idx+1]
# Check the directly right object's color
direct_right_color = direct_right[1]
answer = direct_right_color
```

***Q: Here is a table where the first line is a header and each subsequent line is a penguin: name, age, height (cm), weight (kg) Louis, 7, 50, 11 Bernard, 5, 80, 13 Vincent, 9, 60, 11 Gwen, 8, 70, 15 For example: the age of Louis is 7, the weight of Gwen is 15 kg, the height of Bernard is 80 cm. We now add a penguin to the table: James, 12, 90, 12

How many penguins are less than 8 years old?

```
# Put the penguins into a list.
penguins = []
penguins.append(('Louis', 7, 50, 11))
penguins.append(('Bernard', 5, 80, 13))
penguins.append(('Vincent', 9, 60, 11))
penguins.append(('Gwen', 8, 70, 15))
# Add penguin James.
penguins.append(('James', 12, 90, 12))
# Find penguins under 8 years old.
penguins_under_8_years_old = [penguin for penguin in penguins if penguin[1] < 8]
# Count number of penguins under 8.
num_penguin_under_8 = len(penguins_under_8_years_old)
answer = num_penguin_under_8
```

PoT (Program of Thought), 利用代码可执行性，实现精准的数值、时序、表格、空间推理

Chen, Wenhui, et al. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. *arXiv preprint arXiv:2211.12588* (2022)

Gao, Luyu, et al. Pal: Program-aided language models. *International Conference on Machine Learning*. PMLR, 2023.



大纲

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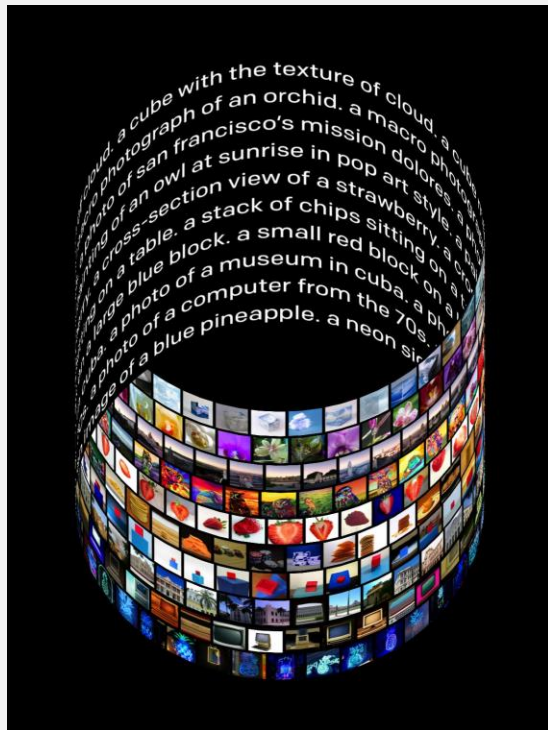
背景

□ 多模态数据

- 语言
- 图像
- 视频

□ 跨模态应用

- 图像描述生成（Image Captioning）
- 跨模态检索（如：以文搜图/视频）
- 辅助单模态任务
-



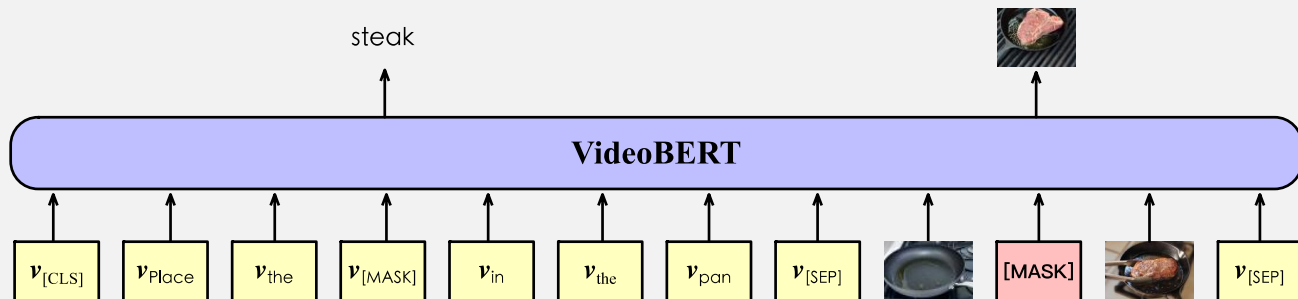
图片来源: <https://openai.com/blog/dall-e/>



掩码图像模型——VideoBERT

□ Videobert: A joint model for video and language representation learning.
(Sun et al., ICCV 2019)

- “文本+视频” 预训练模型
- 数据：视频以及对应的文本字幕
- 预训练任务：掩码标记预测



□ 应用

- 视频检索、视频字幕生成等

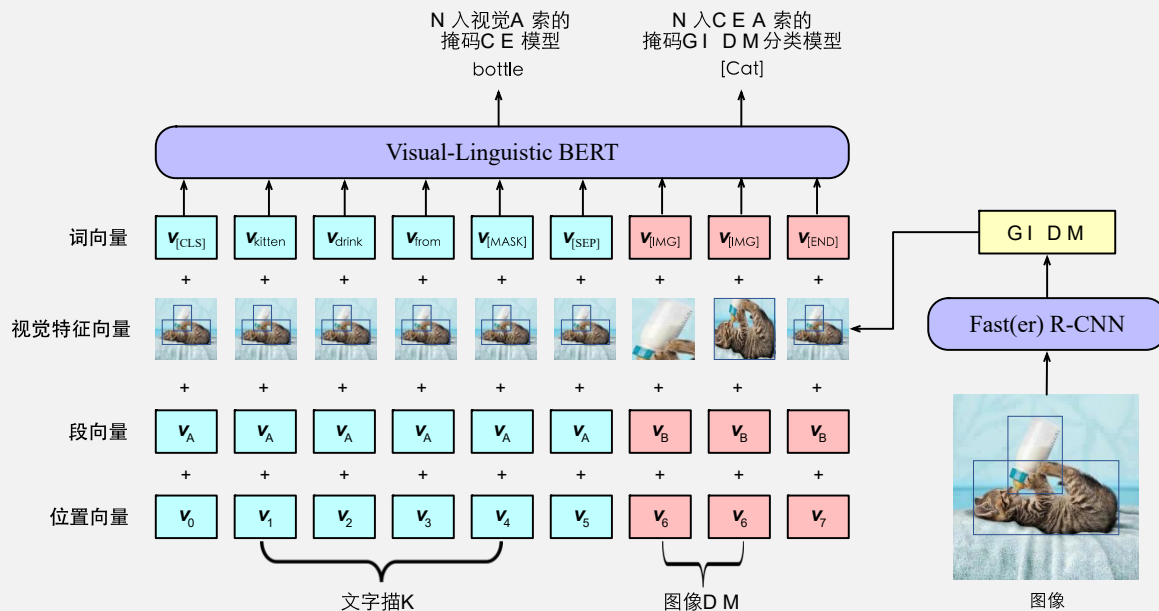




掩码图像模型——VL-BERT

□ VI-bert: Pre-training of generic visual-linguistic representations. (Su et al., ICLR 2019)

– “文本+图像”预训练模型

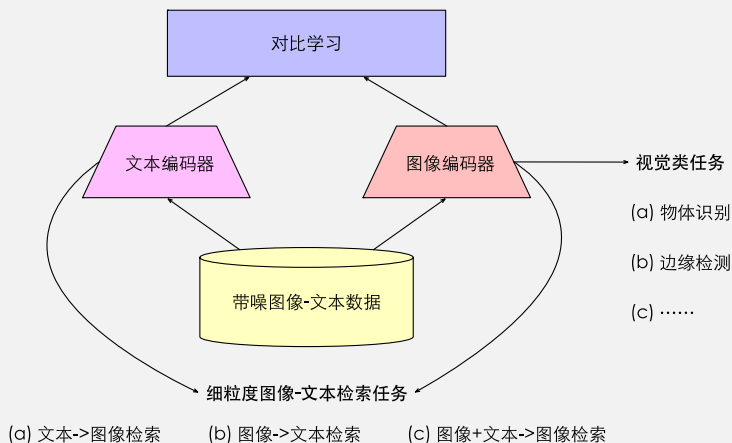




基于对比学习的多模态预训练模型 —— CLIP, ALIGN

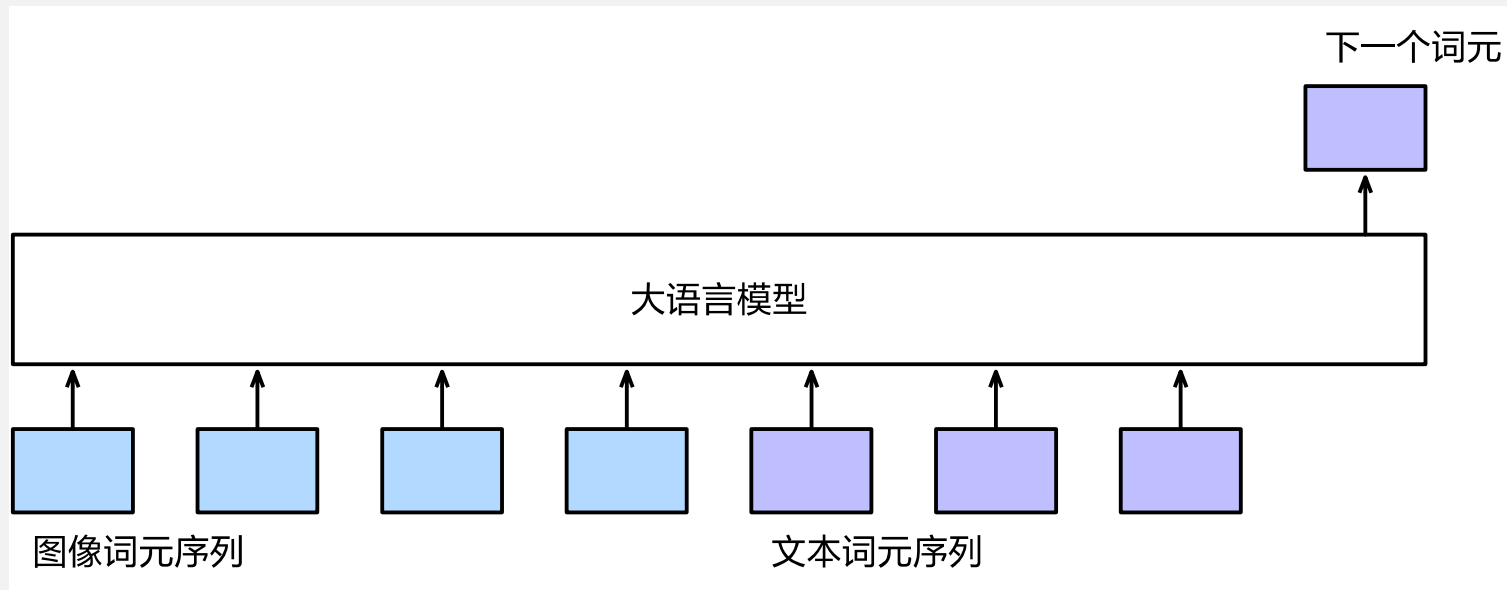
- CLIP: Learning Transferable Visual Models From Natural Language Supervision (Radford, et al., 2021)
- ALIGN: Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision (Jia et al., 2021)
- 通过对比学习学习图像与文本的联合表示

预训练





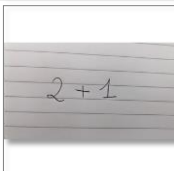
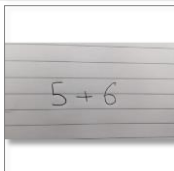
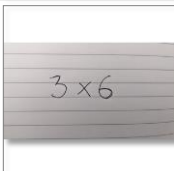
图到文预训练模型



图文模型的预训练过程



图到文预训练模型——Flamingo

输入提示					输出
	This is a chinchilla. They are mainly found in Chile.		This is a shiba. They are very popular in Japan.		This is a flamingo. They are found in the Caribbean and South America.
	What is the title of this painting? Answer: The Hallucinogenic Toreador.		Where is this painting displayed? Answer: Louvres Museum, Paris.		What is the name of the city where this was painted? Answer: Arles.
	Output: "Underground"		Output: "Congress"		Output: "Souloles"
	2+1=3		5+6=11		3x6=18



图像或视频生成——DALL · E

□ Zero-Shot Text-to-Image Generation (Ramesh et al, 2021)

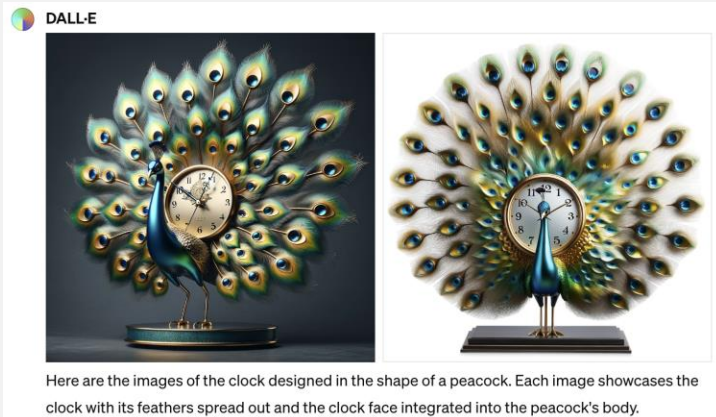
- 通过离散变分自编码器（Discrete VAE）将图像表示为离散标记序列
- 由文本生成图像，采用自回归语言模型

输入：

a clock in the shape of a peacock.

（一个孔雀形的时钟）

输出：





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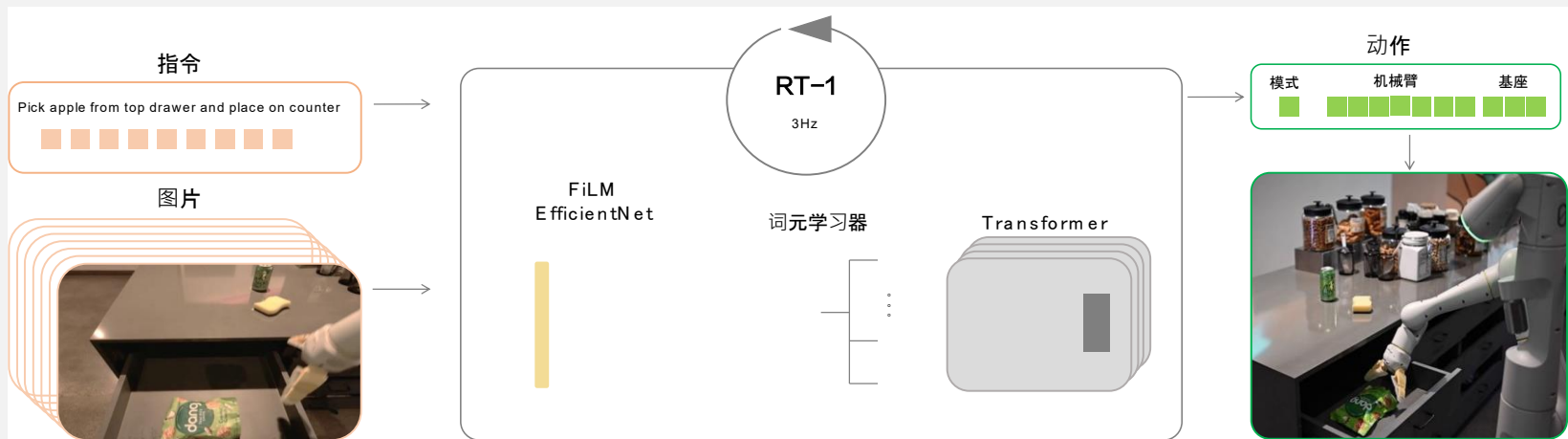


具身智能

- 具身智能（Embodied Artificial Intelligence, Embodied Intelligence 或 Embodied AI）
 - 能够同物理世界进行交互的智能体
- 通过大语言模型，将人类与机器人交互的指令转化为机器人能够执行的具体动作
 - 类似人类大脑执行的操作：将复杂的抽象指令转化为一系列简单的具体指令
 - 类似人类小脑执行的操作：将具体的指令转化为机器人能够实际执行的动作序列



RT-1 (Robotics Transformer)





PaLM-E

移动操控



Human: Bring me the rice chips from the drawer. Robot: 1. Go to the drawers, 2. Open top drawer. I see . 3. Pick the green rice chip bag from the drawer and place it on the counter.

视觉问答、字幕生成、...



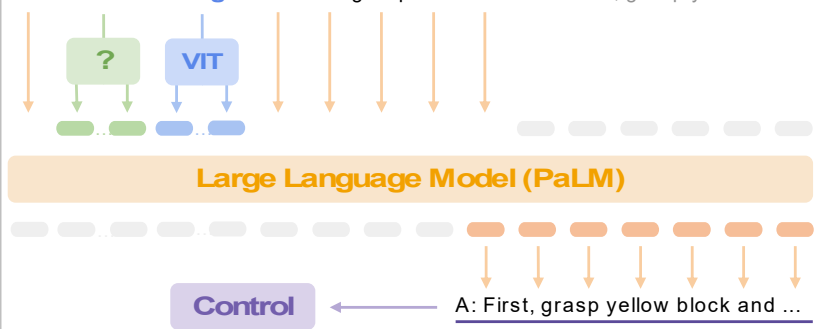
Given . Q: What's in the image? Answer in emojis.
A: 🍏 🍌 🍇 🍎 🍓 🍓



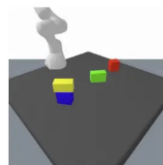
Describe the following :
A dog jumping over a hurdle at a dog show.

PaLM-E: An Embodied Multimodal Language Model

Given ... Q: How to grasp blue block? A: First, grasp yellow block



任务和动作规划



Given Q: How to grasp blue block?
A: First grasp yellow block and place it on the table, then grasp the blue block.

桌面操控



Given Task: Sort colors into corners.
Step 1. Push the green star to the bottom left.
Step 2. Push the green circle to the green star.

仅限语言的任务

Here is a Haiku about embodied language models:
Embodied language models are the future of natural language

Q: Miami Beach borders which ocean? A: Atlantic.
Q: What is 372×18 ? A: 6696.
Language models trained on robot sensor data can be used to guide a robot's actions.



RT-2

互联网规模 VQA + 机器人动作数据



Q: What is happening in the image?

A: 311 423 170 55 244

A grey donkey walks down the street.

Q: Que puis-je faire avec ces objets?

A: 3455 1144 189 25673

Faire cuire un gâteau.



Q: What should the robot do to **<task>**?

A: 132 114 128 5 25 156

$\Delta \text{Translation} = [0.1, -0.2, 0]$
 $\Delta \text{Rotation} = [10, 25, 7]$

协同微调

机器人控制的视觉-语言-动作模型

Q: What should the robot do to **<task>**?

A: ...



RT-2

ViT

Large Language Model

A: 132 114 128 5 25 156

反词元化

$\Delta T = [0.1, -0.2, 0]$
 $\Delta R = [10, 25, -7]$

部署

闭环机器人控制



Put the strawberry into the correct bowl



Pick the nearly falling bag



Pick object that is different



南京大学
NANJING UNIVERSITY



小结

- 多语言预训练模型
- 代码预训练模型
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